

Original Investigation

Vision Loss Following Episcleral Brachytherapy for Uveal Melanoma

Development of a Vision Prognostication Tool

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IMPORTANCE Vision loss following episcleral brachytherapy for uveal melanoma is difficult to predict for individual patients.

OBJECTIVE To generate a risk calculator for vision loss following episcleral brachytherapy for uveal melanoma.

DESIGN, SETTING, AND PARTICIPANTS A retrospective review of data was conducted at a multispecialty tertiary care center in Cleveland, Ohio. All patients with primary ciliary body or choroidal melanoma treated with iodine 125 or ruthenium 106 episcleral brachytherapy between January 1, 2004, and December 30, 2013, were included. Univariate and multivariable Cox proportional hazards were used to determine the influence of baseline patient factors on vision loss. Kaplan-Meier curves (log-rank analyses) were used to estimate freedom from vision loss. Bootstrap resampling was performed to bias correct this estimate.

MAIN OUTCOMES AND MEASURES Vision loss (to visual acuity [VA] worse than 20/50 and worse than 20/200).

RESULTS A total of 311 patients were included in the study, with a mean (SD) age of 62 (14.7) years at start of treatment and a median follow-up of 36 months (interquartile range, 18-60 months). At presentation, VA was better than or equal to 20/50 in 199 patients (64%) and better than or equal to 20/200 in 289 patients (93%). By Kaplan-Meier analysis, VA less than 20/200 at 3 years was not associated with sex, diabetes, systemic hypertension, or hypercholesterolemia but was associated with history of ocular comorbidities, type of isotope (ruthenium 106 or iodine 125), and initial VA ($>20/50$ or $<20/50$). By multivariable analysis, age (hazard ratio [HR], 0.97; 95% CI, 0.94-1.00; $P = .06$), largest basal diameter (HR, 1.25; 95% CI, 1.16-1.34; $P = <.001$), total radiation dose to the fovea (HR, 1.03; 95% CI, 1.01-1.04; $P = .001$) and optic disc (HR, 1.01; 95% CI, 1.00-1.01; $P = .005$), and initial VA worse than 20/50 (HR, 1.85; 95% CI, 1.20-2.85; $P = .005$) were predictive of vision loss to a VA of less than 20/200. The concordance index for the full data set was 0.77. Using these data, an online risk calculator was developed to predict vision loss following episcleral brachytherapy.

CONCLUSIONS AND RELEVANCE The vision prognostication tool presented herein needs to be validated by independent data sets. This tool may improve counseling for patients being evaluated for episcleral brachytherapy. At-risk individuals identified by this tool could be considered for inclusion into trials exploring prevention or treatment of radiation retinopathy and alternative therapies of uveal melanoma.

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The Collaborative Ocular Melanoma Study firmly established iodine 125 (^{125}I) episcleral brachytherapy as an alternative treatment to enucleation for medium-sized choroidal melanomas.^{1,2} However, this globe-preserving treatment modality leads to vision loss in most patients. The Collaborative Ocular Melanoma Study reported that at 3 years, only 43% of patients would have visual acuity (VA) better than 20/200.²

Several subsequent studies have attempted to elucidate the rate of and associated factors contributing to the decline in vision following episcleral brachytherapy with radioactive isotopes ^{125}I , ruthenium 106 (^{106}Ru), and palladium 103 (Table 1).²⁻¹⁹ Some of those factors are inherent characteristics of the tumor, such as tumor size and its proximity to the fovea. Other factors include secondary effects of radiation on normal ocular structures, including radiation retinopathy (ischemic, exudative, hemorrhagic, or atrophic), radiation chorioidopathy, radiation optic neuropathy, and radiation-induced cataract.²⁰ Secondary complications, such as vitreous hemorrhage or toxic tumor syndrome, may also contribute to vision loss.²¹

From the patient's perspective, knowledge of visual outcome following episcleral brachytherapy is an important

Key Points

Question Is it possible to offer individualized vision prognostication for patients undergoing episcleral brachytherapy for uveal melanoma?

Findings In this study of vision outcomes of 311 patients following episcleral brachytherapy, predictors for vision loss were identified, and a vision prognostication model was developed, refined, and internally validated.

Meaning Following validation, this online vision prognostication tool might be used to predict vision loss based on parameters available prior to surgical implantation of the episcleral plaque.

consideration for choosing conservative therapy over enucleation. Because factors resulting in vision loss after radiation do not occur in isolation, determining the specific etiopathogenetic mechanism for vision loss may be difficult to ascertain in an individual patient.^{20,21} Therefore, determining the overall risk of vision loss based on individual tumor characteristics and the presence of ocular and systemic diseases may help in patient selection and management counseling.

Table 1. Published Studies Evaluating VA After Plaque Brachytherapy for Uveal Melanoma^a

Source	Isotope	Patients, No.	Median Follow-up, mo	Patients With VA Worse Than 20/200, %	Factors Associated With Decreased VA
Sia et al, ³ 2000	^{125}I	49	39	37	No predictors found
Melia et al, ² 2001	^{125}I	623	...	45	Tumor-associated retinal detachment; tumor not dome shaped; diabetes
Jones et al, ⁴ 2002	^{125}I	63	36	...	Tumor proximity to macula and optic disc; higher doses to macula and optic disc; macula dose rate, mean (SD), 111 (11.1) cGy/h
Shields et al, ⁵ 2002	^{125}I	354	120	89	Increased age; retinal invasion by melanoma; <2-mm distance from the optic disc
Bechrakis et al, ⁶ 2002	^{125}I	152	27 ^b	95	Greater tumor height; increased age
Finger et al, ⁷ 2002	^{103}Pd	152	55	27	Posterior location of the tumor
Puusaari et al, ⁸ 2004	^{125}I	96	42	18 ^c	...
Rouberol et al, ⁹ 2004	^{106}Ru	213	...	...	Tumor location
Stack et al, ¹⁰ 2005	^{125}I	92	31	49	...
Jensen et al, ¹¹ 2005	^{125}I	156	74	56	Dose rate to optic disc >60 cGy/h; dose rate to lens <15 cGy/h; atherosclerosis or coronary artery disease
Bergman et al, ¹² 2005	^{106}Ru	579	82	51	Initial VA; distance to the fovea
Damato et al, ¹³ 2005	^{106}Ru	458	60	43	Reduced initial VA; posterior tumor extension; temporal tumor location
Krohn et al, ¹⁴ 2008	^{125}I	111	61	60	Tumor height >8 mm; large tumor
Saconn et al, ¹⁵ 2010	^{125}I	62	58	22	Diabetic retinopathy; poor initial VA in affected eye
Verschueren et al, ¹⁶ 2010	^{106}Ru	425	50	48	Initial VA; tumor height; tumor location
Khan et al, ¹⁷ 2012	^{125}I	189	...	25	Greater total dose to fovea; greater tumor height; increased age
Perez et al, ¹⁸ 2014	^{125}I	190	49	...	Lower radiation dose (<85 Gy) to the tumor apex
Wisely et al, ¹⁹ 2016	^{125}I	113	66	42	Baseline VA; apical tumor height; AJCC tumor category; distance to fovea
Current study	^{125}I , ^{106}Ru	311	36	15 (1y); 27 (3y) ^d	Age; largest basal dimension; surface area; dose to fovea; dose to optic disc; initial VA

Abbreviations: AJCC, American Joint Committee on Cancer; ellipses, data unknown; ^{125}I , iodine 125; ^{103}Pd , palladium 103; ^{106}Ru , ruthenium 106; VA, visual acuity.

^a Includes English-language studies published between 2000 and 2015 evaluating VA after plaque brachytherapy for uveal melanoma.

^b Value expressed as mean.

^c Of patients with VA better than 20/400.

^d Of the 289 at risk to VA of 20/200 vision loss at baseline.

This study is an extension of our preliminary work¹⁷ and includes a larger number of patients with a longer follow-up and the creation of an online risk calculator.

Methods

Patients treated at the Cole Eye Institute at the Cleveland Clinic in Ohio for a primary single ciliary body or choroid melanoma with ¹²⁵I or ¹⁰⁶Ru episcleral brachytherapy between January 1, 2004, and December 30, 2013, were included. Exclusion criteria included patients with previous radiation treatment and patients who received adjuvant transpupillary thermotherapy. Written approval for exempt status was obtained from the institutional review board at the Cleveland Clinic Foundation for this study because anonymous data were collected and analyzed.

Patient demographic characteristics and clinical information were extracted from the Cleveland Clinic electronic medical records system (EpicCare EMR; Epic Systems Corp) and paper medical records. Dosimetry for the plaques was performed with plaque simulator treatment planning software (Plaque Simulator version 5.3.4; Bebig GmbH) designed to give a total prescription of 85 Gy to the tumor apex. Isotopes used were ¹²⁵I (model MED3631-A/M; North American Scientific Inc; or Isoaid Advantage model IAI-125A; Isoaid LLC) and ¹⁰⁶Ru (models CCA, CCD, and CCB, with diameters of 15.3 mm, 17.9 mm, and 20.2 mm, respectively; Bebig GmbH).

As part of the initial workup, all patients underwent a detailed ophthalmoscopic examination, photography, and ultrasonographic A and B scans. Tumor size (basal dimensions and height) was assessed by ophthalmoscopic examination and by ultrasonography. Tumor localization for plaque placement was performed by use of transillumination or indirect ophthalmoscopy, based on tumor location. Intraoperative ultrasonography was performed to ensure that all tumor margins were covered by the plaque. The patients were evaluated for follow-up 1 week after episcleral brachytherapy, then every 3 months throughout the first year, every 6 months throughout the second through fifth years, and annually thereafter. The best-corrected VA using a Snellen eye chart was measured at each visit (converted to logMAR values for statistical analysis).

Exploratory frequency tables and descriptive statistics were created for the outcomes of VA worse than 20/50 and VA worse than 20/200. Categorical factors were described using frequencies and percentages, while continuous measures were summarized as means and SDs. Comparisons were made using Pearson χ^2 tests or 2-sample *t* tests as appropriate.

Kaplan-Meier estimation and Cox proportional hazards models were performed to identify potential predictors of VA loss. Because the number of events did not permit inclusion of all these factors, data reduction was performed by choosing the factors that best approximated the fit of the full model with the least loss of information. This model was used as the final model. Restricted cubic splines were used for continuous measures to allow for nonlinear relationships with risk of vision loss. After this model was fitted, the ability of the model

to discriminate those with vision loss from those without vision loss was measured using a C statistic. The concordance index reflects the proportion of cases where those with worse vision outcome were assigned higher risk by the model compared with those with better vision. A C statistic of 0.5 indicates a random guess, while a C statistic of 1 indicates perfect discrimination. Bootstrap resampling was performed to bias correct this estimate.²² Similar methods were used to evaluate the calibration of the model (agreement between predicted and actual risk). Here, a calibration plot reflects the predicted likelihood of vision at 1 year vs the actual observed vision. The quality of the calibration of actual and model-predicted levels is evaluated by how closely the 2 measures equal one another, as depicted by the line of equality in the plot. Predicted risk was calculated at 1 year and 3 years. Analyses were performed using functions from within R software version 3.0 (R Foundation). A nomogram of the findings was created. Based on the nomogram data, an online risk calculator for vision loss was developed (eFigure in the [Supplement](#) and http://riskcalc.org:3838/3-year_Risk_of_Visual_Acuity/).

Results

Between January 1, 2004, and December 30, 2013, 353 cases of choroidal melanoma and/or ciliary body were treated with episcleral brachytherapy at the Cole Eye Institute at the Cleveland Clinic. Of these, 42 patients were excluded: 36 patients had adjuvant transpupillary thermotherapy and 6 patients had previous radiation treatment. The remaining 311 cases of ciliary body and/or choroidal melanoma were included in this study. The mean (SD) age of included participants was 62 (14.7) years at the start of treatment. The median follow-up was 36 months (interquartile range, 18-60).

Visual acuity was evaluated, excluding all patients with VA worse than the specified end points at presentation. Therefore, 199 patients with VA better than or equal to 20/50 at presentation were included in the analysis for factors associated with VA worse than 20/50, and 289 patients with VA better than or equal to 20/200 were included in the analysis for factors associated with VA worse than 20/200. The demographic characteristics of the patients, overall tumor characteristics, and radiation dose distribution are summarized in [Table 2](#).

Factors associated with vision loss on univariate analysis with Cox proportional hazards models are presented in [eTable 1](#) (noncategorical factors) and [eTable 2](#) and [eTable 3](#) in the [Supplement](#) (categorical factors). Among noncategorical factors, tumor size (largest basal diameter, surface area, and height of tumor), tumor location (in millimeters; distance to fovea and optic disc), and total radiation dose (in grays; tumor apex, fovea, lens, and disc) had a *P* value <.05 for VA worse than 20/200 ([eTable 1](#) in the [Supplement](#)). Of these, an association was not identified between distance to fovea and optic disc and total dose to apex for VA worse than 20/50.

Kaplan-Meier estimates of associations of categorical factors with time to VA worse than 20/50 revealed ocular comorbidities (cataract, glaucoma, macular degeneration, strabis-

Table 2. Patient Demographic Characteristics, Tumor Characteristics, and Radiation Dose Distribution

Characteristic	No. (%)
Total patients	311
Age at start of treatment, mean (SD), y	62 (14.7)
Follow-up time, median (IQR), mo	36 (18, 60)
Sex	
Female	155 (49.8)
Male	156 (50.2)
Diabetes	49 (15.8)
Hypertension	128 (41.2)
Heart disease	87 (28.0)
Ocular comorbidities ^a	139 (44.7)
Tumor, mean (SD)	
Tumor height, mm	4.0 (2.2)
Largest basal diameter, mm	11.1 (3.2)
Distance to fovea, mm	5.1 (4.5)
Distance to optic disc, mm	5.7 (3.9)
Surface area, mm ²	100.3 (61.5)
Radiation, mean (SD), Gy	
Total dose to apex	85.9 (8.3)
Total dose to fovea	58.4 (52.2)
Total dose to lens	15.2 (19.0)
Total dose to disc	31.5 (27.4)
Isotope	
Iodine 125	271 (87.1)
Ruthenium 106	40 (12.9)

Abbreviation: IQR, interquartile range.

^a Ocular comorbidities include cataract, glaucoma, macular degeneration, strabismus, amblyopia, retinal disease, history of injury, and surgery.

mus, amblyopia, retinal disease, history of injury, and surgery) to be associated ($P = .02$), whereas an association was not identified with systemic conditions such as diabetes ($P = .36$), hypertension ($P = .49$), or hypercholesterolemia ($P = .32$) or with the type of radioactive isotope used ($P = .36$ for all) (eTable 2 in the Supplement). However, for time to VA better than or equal to 20/200, associations were observed with the type of radioactive isotope used (^{125}I or ^{106}Ru ; $P = .003$) and initial VA (20/50 or worse; $P = .001$) (eTable 3 in the Supplement).

On multivariable analysis for VA worse than 20/200, factors shown to be detrimental to VA included age at start of treatment (hazard ratio [HR], 0.97; 95% CI, 0.94-1.00; $P = .06$), largest basal diameter (HR, 1.25; 95% CI, 1.16-1.34; $P < .001$), total dose to fovea (HR, 1.03; 95% CI, 1.01-1.04; $P = .001$) and optic disc (HR, 1.01; 95% CI, 1.00-1.01; $P = .005$), and initial VA (20/50 or worse; HR, 1.85; 95% CI, 1.20-2.85; $P = .005$) (Table 3). Of note, type of radioactive isotope used (^{125}I or ^{106}Ru) was not an independent associated variable.

After performing data reduction by choosing the factors that best approximated the fit of the full model with the least loss of information, a nomogram to predict risk of VA worse than 20/200 at 1 and 3 years of follow-up was developed (Figure 1). An online calculator was developed based on the nomogram, which assigns a certain number of points to statistically significant factors associated with vision loss found

Table 3. Final Model of Predictive Factors for Visual Acuity Worse Than 20/200

Factor	Hazard Ratio (95% CI) ^a	P Value
Age at start of treatment	0.97 (0.94-1.00)	.06
Largest basal diameter, mm	1.25 (1.16-1.34)	<.001
Total dose to fovea, Gy	1.03 (1.01-1.04)	.001
Total dose to optic disc, Gy	1.01 (1.00-1.01)	.005
Initial visual acuity 20/50 or worse, yes vs no	1.85 (1.20-2.85)	.005

^a Hazard ratios come from Cox proportional hazard models.

on multivariable analysis (eFigure in the Supplement and http://riskcalc.org:3838/3-year_Risk_of_Visual_Acuity/). The ability of the model to discriminate those with vision loss from those without vision loss was measured using a concordance index (0.77) between actual and predicted vision loss (Figure 2). A nomogram could not be generated for VA worse than 20/50 because it was not possible to fit a statistically significant model with adequate discrimination or calibration.

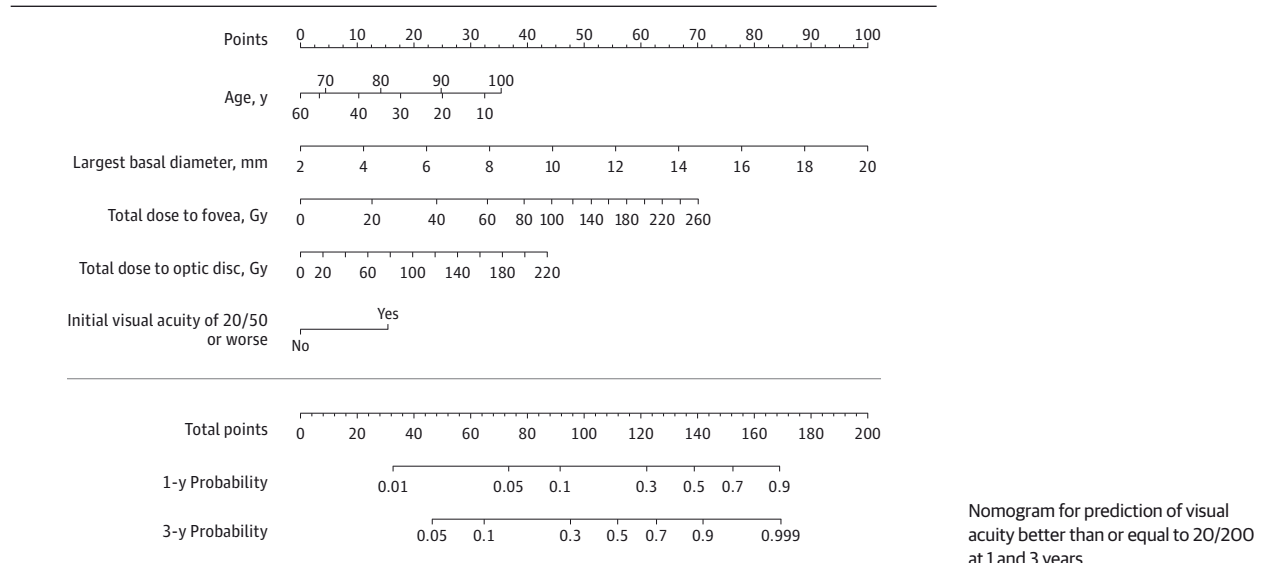
Discussion

Similar to previous studies, our updated data set demonstrates strong associations between decrease in VA and age at presentation, initial VA, tumor dimensions, tumor location, and radiation dose to the optic disc and fovea.^{4-7,9,12-17} Systemic factors (eg, diabetes and hypertension) have not been consistently reported as risk factors in predicting final VA (Table 1).^{2,11,15} In our data set, diabetes was not identified as a risk factor (15.8% of patients had diabetes).

Our data also revealed that the main modifiable factor associated with VA worse than 20/200 was the total radiation dose to the fovea and the optic disc. Therefore, if the radiation dose could be reduced, visual outcome after brachytherapy could potentially be improved. Sacconn et al¹⁵ have shown equivalent control rates with a dose of 63 Gy to the tumor apex while preserving VA better than 20/200 in up to 78% of patients at 5-year follow up. A larger prospective study of dose de-escalation is required to identify the threshold doses necessary to achieve local tumor control while minimizing the risk of radiation-induced vision loss.

Although our data set is based on 2 types of radioactive isotopes (^{125}I and ^{106}Ru), the type of isotope used was not an associated independent variable in the final model (Table 3). Similarly, the distance of the tumor from the fovea and optic disc was not an associated independent variable. In the multivariable analysis for VA worse than 20/200, the total radiation dose to the fovea (HR, 1.03; 95% CI, 1.01-1.04; $P = .001$) and the optic disc (HR, 1.01; 95% CI, 1.00-1.01; $P = .005$) appeared to be associated. Total radiation dose to the fovea and the optic disc are influenced by the distance between the tumor and these structures and also by the isotope-dependent isodose distribution, which in our model were reflected in the total radiation dose. Perhaps in larger data sets, subtle influences of the type of isotope used and distance to fovea and optic disc on VA may become detectable.

Figure 1. Nomogram for Prediction of Visual Acuity



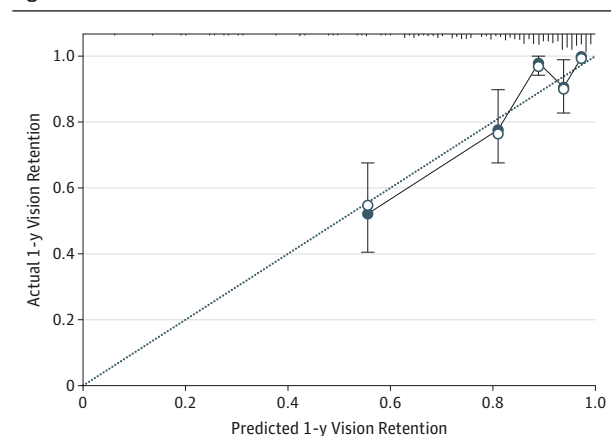
The limitations of our data include the retrospective collection of clinical data and the relatively small number of patients. Therefore, it is possible that some of the prognostic factors may have been excluded because they did not achieve adequate statistical significance to be included in the model. On the other hand, the strengths of this study's data include continued and uniform follow-up patterns on most of the patients. Additionally, the use of a standard radiation treatment planning protocol with validated software provided comparable and reproducible data throughout the study duration.

The methods for evaluating predictive models differ primarily in the validation aspects of the model and focus less on interpretation of the associated individual factors. We recognize that the model presented herein would need to be externally validated to evaluate it, but the purpose of our study was to create and report a model that could be evaluated. While external validation is one way to evaluate a model, it is not the only method. As noted in the Methods section, internal validation (bootstrap resampling) was performed to evaluate the model performance. These methods have shown similar performance to split-sample validation methods.²² The ability of the final model to discriminate those with vision loss from those without vision loss was measured using a C statistic (concordance index, 0.77). These statistics reflect that approximately 77% of the time, those with vision loss were correctly identified by the model. Actual and predicted risk levels are generally well matched, except at the very high end of distribution, where predicted vision retention exceeds actual vision retention. Ideally, the plots of predicted and actual risk measures would be close to the "ideal" line (Figure 2).

Conclusions

All needed parameters to predict vision loss based on patient characteristics, tumor characteristics, and dosimetry are available prior to surgical implantation of the episcleral plaque. Once validated, the risk calculator can assist clinicians and pa-

Figure 2. Actual vs Predicted 1-Year Vision Retention



The calibration plot for visual acuity better than or equal to 20/200 (sample size = 289; 100 patients with vision loss; 7 parameters in the model; 1000 bootstrap sample). Ideally, the plots of predicted and actual risk measures would be close to the "ideal" line (black line). The concordance index for the full data set was 0.77. These statistics reflect that approximately 77% of the time, those with vision loss were correctly identified by the model. The steady increase in the points on the calibration curve indicate that actual and predicted risk levels are generally well matched, except at the very high end of distribution, where predicted vision retention exceeds the actual vision retention. Error bars indicate 95% CIs. Filled points indicate model-predicted vision retention rates; empty points indicate bootstrap (optimism)-adjusted vision retention rates. Tick marks at the top of the graph indicate the distribution of participants.

tients make an educated decision on treatment options based on the patient-specific risk of vision loss rather than global risk of vision loss. Additionally, the risk calculator can be used to assess new isotopes or designs for ophthalmic episcleral brachytherapy implants. At-risk individuals identified by this tool could be considered for inclusion into trials exploring prevention or treatment of radiation retinopathy and alternative therapies for uveal melanoma.

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Acquisition, analysis, or interpretation of data: Aziz, N. Singh, Bena, Wilkinson.

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REFERENCES

- Krintz AL, Hanson WF, Ibbott GS, Followill DS. A reanalysis of the Collaborative Ocular Melanoma Study Medium Tumor Trial eye plaque dosimetry. *Int J Radiat Oncol Biol Phys*. 2003;56(3):889-898.
- Melia BM, Abramson DH, Albert DM, et al; Collaborative Ocular Melanoma Study Group. Collaborative Ocular Melanoma Study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma, I: visual acuity after 3 years COMS report No. 16. *Ophthalmology*. 2001;108(2):348-366.
- Sia S, Harper C, McAllister I, Perry A. Iodine-125 episcleral plaque therapy in uveal melanoma. *Clin Experiment Ophthalmol*. 2000;28(6):409-413.
- Jones R, Gore E, Mieler W, et al. Posttreatment visual acuity in patients treated with episcleral plaque therapy for choroidal melanomas: dose and dose rate effects. *Int J Radiat Oncol Biol Phys*. 2002;52(4):989-995.
- Shields CL, Naseripour M, Cater J, et al. Plaque radiotherapy for large posterior uveal melanomas (> or =8-mm thick) in 354 consecutive patients. *Ophthalmology*. 2002;109(10):1838-1849.
- Bechrakis NE, Bornfeld N, Zöller I, Foerster MH. Iodine 125 plaque brachytherapy versus transscleral tumor resection in the treatment of large uveal melanomas. *Ophthalmology*. 2002;109(10):1855-1861.
- Finger PT, Berson A, Ng T, Szechter A. Palladium-103 plaque radiotherapy for choroidal melanoma: an 11-year study. *Int J Radiat Oncol Biol Phys*. 2002;54(5):1438-1445.
- Puusaari I, Heikkonen J, Kivelä T. Effect of radiation dose on ocular complications after iodine brachytherapy for large uveal melanoma: empirical data and simulation of collimating plaques. *Invest Ophthalmol Vis Sci*. 2004;45(10):3425-3434.
- Rouberol F, Roy P, Kodjikian L, Gérard JP, Jean-Louis B, Grange JD. Survival, anatomic, and functional long-term results in choroidal and ciliary body melanoma after ruthenium brachytherapy (15 years' experience with beta-rays). *Am J Ophthalmol*. 2004;137(5):893-900.
- Stack R, Elder M, Abdelaal A, Hidajat R, Clemett R. New Zealand experience of I125 brachytherapy for choroidal melanoma. *Clin Experiment Ophthalmol*. 2005;33(5):490-494.
- Jensen AW, Petersen IA, Kline RW, Stafford SL, Schomberg PJ, Robertson DM. Radiation complications and tumor control after I251 plaque brachytherapy for ocular melanoma. *Int J Radiat Oncol Biol Phys*. 2005;63(1):101-108.
- Bergman L, Nilsson B, Lundell G, Lundell M, Seregard S. Ruthenium brachytherapy for uveal melanoma, 1979-2003: survival and functional outcomes in the Swedish population. *Ophthalmology*. 2005;112(5):834-840.
- Damato B, Patel I, Campbell IR, Mayles HM, Errington RD. Visual acuity after Ruthenium(106) brachytherapy of choroidal melanomas. *Int J Radiat Oncol Biol Phys*. 2005;63(2):392-400.
- Krohn J, Monge OR, Skorpen TN, Mørk SJ, Dahl O. Posterior uveal melanoma treated with I-125 brachytherapy or primary enucleation. *Eye (Lond)*. 2008;22(11):1398-1403.
- Saccon PA, Gee CJ, Greven CM, McCoy TP, Ekstrand KE, Greven KM. Alternative dose for choroidal melanoma treated with an iodine-125 radioactive plaque: a single-institution retrospective study. *Int J Radiat Oncol Biol Phys*. 2010;78(3):844-848.
- Verschueren KM, Creutzberg CL, Schalijs-Delfos NE, et al. Long-term outcomes of eye-conserving treatment with Ruthenium(106) brachytherapy for choroidal melanoma. *Radiother Oncol*. 2010;95(3):332-338.
- Khan N, Khan MK, Bena J, Macklis R, Singh AD. Plaque brachytherapy for uveal melanoma: a vision prognostication model. *Int J Radiat Oncol Biol Phys*. 2012;84(3):e285-e290. doi:10.1016/j.ijrobp.2012.04.005.
- Perez BA, Mettu P, Vajzovic L, et al. Uveal melanoma treated with iodine-125 episcleral plaque: an analysis of dose on disease control and visual outcomes. *Int J Radiat Oncol Biol Phys*. 2014;89(1):127-136.
- Wisely CE, Hadziahmetovic M, Reem RE, et al. Long-term visual acuity outcomes in patients with uveal melanoma treated with (125)I episcleral OSU-Nag plaque brachytherapy. *Brachytherapy*. 2016;15(1):12-22.
- Aziz HA, Singh AD. Radiation chorioretinopathy: evidence for effective therapy. *Int Ophthalmol Clin*. 2015;55(1):53-61.
- Singh AD, Pabon S, Aronow ME. Management of radiation maculopathy. *Ophthalmic Res*. 2012;48(suppl 1):26-31.
- Steyerberg EW, Harrell FE Jr, Borsboom GJ, Eijkemans MJ, Vergouwe Y, Habbema JD. Internal validation of predictive models: efficiency of some procedures for logistic regression analysis. *J Clin Epidemiol*. 2001;54(8):774-781.